

$$(\vec{E}, \varepsilon, \pi) \quad \pi: \begin{array}{ccc} \varepsilon \times \varepsilon & \longrightarrow & \vec{\varepsilon} \\ (p, q) & \longmapsto & \overrightarrow{pq} \end{array}$$

Ricordiamo: $\mathcal{H} \subseteq \varepsilon$ è sottospazio euclideo (o affine) se

(i) $\vec{\mathcal{H}} = \pi(\mathcal{H} \times \mathcal{H}) = \{ \overrightarrow{pq} \mid p, q \in \mathcal{H} \}$ è sottosp. vett. di $\vec{\varepsilon}$

(ii) $\forall A \in \mathcal{H}, \forall a \in \vec{\mathcal{H}}$, l'unico punto $X \in \varepsilon$ tale che $\overrightarrow{AX} = a$ appartiene ad $\mathcal{H}$.

$\forall p_0 \in \varepsilon, \forall U \subseteq \vec{\varepsilon}$ sottosp. vett. possiamo considerare:

$$(p_0, U) \stackrel{\text{def.}}{=} \{ q \in \varepsilon \mid \overrightarrow{p_0 q} \in U \} \quad \text{varietà-lineare}$$

$$p_0 + U$$

• Ogni varietà-lineare è un sottosp. euclideo (o affine)

DM: Th: (p_0, U) soddisfa le condiz. (i) e (ii)

(FACOLTATIVA)

$$- p_0: \overrightarrow{p_0 p_0} = \vec{0} \in U \Rightarrow p_0 \in (p_0, U)$$

per def. di (p_0, U)

- Vediamo che (p_0, U) soddisfa la (ii)

$$A \in (p_0, U), a \in U \quad \text{Sappiamo che } \exists! X \in \varepsilon: \overrightarrow{AX} = a \in U \Rightarrow X \in (p_0, U)$$

- Resta da dimostrare che $U = \pi((p_0, U) \times (p_0, U))$

$$" \supseteq " \quad p, q \in (p_0, U) \Rightarrow \overrightarrow{p_0 p}, \overrightarrow{p_0 q} \in U \Rightarrow -\overrightarrow{p_0 p}, \overrightarrow{p_0 q} \in U \Rightarrow \overrightarrow{p_0}, \overrightarrow{p_0 q} \in U$$

$\nwarrow U$ è sottosp. vett.

$$\text{Allora: } \overrightarrow{pq} = \overrightarrow{p_0 p} + \overrightarrow{p_0 q} \in U$$

$" \pi(p, q) "$

$$" \subseteq " \quad u \in U, p_0 \in (p_0, U) \text{ e vale (ii)} \Rightarrow \exists! X \in (p_0, U): \overrightarrow{p_0 X} = u \in U$$

$\pi((p, X))$

• Ogni sottosp. euclideo (o affine) è una varietà-lineare:

DM: $(\vec{\mathcal{H}}, \mathcal{H}, \pi)$ Th: $\forall p_0 \in \mathcal{H}, \mathcal{H} = (p_0, \vec{\mathcal{H}})$ (FACOLTATIVA)

$$" \subseteq " \quad p \in \mathcal{H}, \quad \vec{\mathcal{H}} = \pi(\mathcal{H} \times \mathcal{H}) = \{ \overrightarrow{pq} \mid p, q \in \mathcal{H} \}$$

$$p_0 \in \mathcal{H} \Rightarrow \overrightarrow{p_0 p} \in \vec{\mathcal{H}} \Rightarrow p \in (p_0, \vec{\mathcal{H}})$$

$$" \supseteq " \quad p \in (p_0, \vec{\mathcal{H}}) \Rightarrow \overrightarrow{p_0 p} \in \vec{\mathcal{H}} \xrightarrow{(ii)} \exists! X \in \mathcal{H}: \overrightarrow{p_0 X} = \overrightarrow{p_0 p}$$

$p_0 \in \mathcal{H}$

$\Rightarrow p = X \in \mathcal{H}$

Ricordiamo di abbiamo come rappresent. i sottosp. euclde. (o affini) di spaz. euclde. di dimensione finita, fissato un rif. cartesiano.

$$(\vec{E}, \varepsilon, \pi) \quad \dim \varepsilon = n \quad R = (O, B)$$

$$\mathcal{H} \text{ sottosp. eucldeo, } \dim \mathcal{H} = h$$

$$" (p_0, \vec{\mathcal{H}}) \quad Q \equiv_R (x_1, \dots, x_n) \in \varepsilon, \quad p_0 \equiv_R (a_1, \dots, a_n)$$

$$Q \in \mathcal{H} \Leftrightarrow \overrightarrow{P_0 Q} \in \overrightarrow{\mathcal{H}} \Leftrightarrow \underset{=(x_1 - a_1, \dots, x_n - a_n)}{\phi_B(\overrightarrow{P_0 Q})} \in \phi_B(\overrightarrow{\mathcal{H}}) : \underline{AX = 0} \quad \text{rang}(A) = n - h.$$

$$\Leftrightarrow A \begin{pmatrix} x_1 - a_1 \\ \vdots \\ x_n - a_n \end{pmatrix} = 0 \quad \Leftrightarrow A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \underbrace{A \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}}_{\underline{b}}$$

$\Sigma: AX = \underline{b}$ ha come soluzioni le n -uple delle coordinate di tutti e soli i punti di $\mathcal{H}$.

$\Sigma_0: AX = 0$ rappresenta $\overrightarrow{\mathcal{H}}$ in B

Sia $B_{\overrightarrow{\mathcal{H}}} = \{v_1, \dots, v_h\}$ una base di $\overrightarrow{\mathcal{H}}$: $\overrightarrow{\mathcal{H}} = \mathcal{L}(v_1, \dots, v_h)$

$$\phi_B(v_1) = (a_{11}^1, \dots, a_{n1}^1)$$

$$\vdots$$

$$\phi_B(v_h) = (a_{1h}^1, \dots, a_{nh}^1)$$

$$Q \in \mathcal{H} \Leftrightarrow \overrightarrow{P_0 Q} \in \overrightarrow{\mathcal{H}} \Leftrightarrow \phi_B(\overrightarrow{P_0 Q}) \in \phi_B(\overrightarrow{\mathcal{H}}) = \mathcal{L}(\phi_B(v_1), \dots, \phi_B(v_h))$$

$$\Leftrightarrow \exists t_1, \dots, t_h \in K : (x_1 - a_1, \dots, x_n - a_n) = t_1(a_{11}^1, \dots, a_{n1}^1) + \dots + t_h(a_{1h}^1, \dots, a_{nh}^1)$$

$$\begin{cases} x_1 = a_1 + t_1 a_{11}^1 + \dots + t_h a_{1h}^1 \\ \vdots \\ x_n = a_n + t_1 a_{n1}^1 + \dots + t_h a_{nh}^1 \end{cases}$$

$t_1, \dots, t_h \in K$
rappresentazione parametrica di $\mathcal{H}$
in R

$\mathcal{H}, \mathcal{H}'$ sottosp. encl. (o affini)

$\mathcal{H}$ e $\mathcal{H}'$ sono paralleli $\Leftrightarrow \overrightarrow{\mathcal{H}} \subseteq \overrightarrow{\mathcal{H}'}$ oppure $\overrightarrow{\mathcal{H}'} \subseteq \overrightarrow{\mathcal{H}}$



Esercizio: $(E, \mathcal{E}, \pi)$ $\dim E = 3$ $R = (O, B)$ $Q \equiv_R (x_1, x_2, x_3)$

Rappresentare in R il piano $\mathcal{H}$ passante per $P_0(3, 4, 2)$ e di equazione

$$\overrightarrow{\mathcal{H}} = \mathcal{L}(v_1(1, 2, 0), v_2(2, 1, 1))$$

$$\mathcal{H}: \begin{cases} x_1 = 3 + t_1 \cdot 1 + t_2 \cdot 2 \\ x_2 = 4 + t_1 \cdot 2 + t_2 \cdot 1 \\ x_3 = 2 + t_1 \cdot 0 + t_2 \cdot 1 \end{cases}$$

$$\begin{cases} x_1 = 3 + t_1 + 2t_2 \\ x_2 = 4 + 2t_1 + t_2 \\ x_3 = 2 + t_2 \end{cases}$$

$$Q \in \mathcal{H} \Leftrightarrow \text{rang} \begin{pmatrix} 1 & 2 & x_1 - 3 \\ 2 & 1 & x_2 - 4 \\ 0 & 1 & x_3 - 2 \end{pmatrix} = 2$$

$$\text{GAUSS: } \underline{a}_2 \rightarrow \underline{a}_2 - 2\underline{a}_1 \quad \begin{pmatrix} 1 & 2 & x_1 - 3 \\ 0 & -3 & x_2 - 4 - 2x_1 + 6 \\ 0 & 1 & x_3 - 2 \end{pmatrix} \quad \underline{a}_3 \rightarrow \underline{a}_3 + \frac{1}{3}\underline{a}_2$$

$$\begin{pmatrix} 1 & 2 & x_1 - 3 \\ 0 & -3 & -2x_1 + x_2 + 2 \\ & & \dots \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & x_4 - 3 \\ 0 & -3 & -2x_4 + x_2 + 2 \\ 0 & 0 & x_3 - 2 - \frac{2}{3}x_4 + \frac{1}{3}x_2 + \frac{2}{3} \end{pmatrix}$$

$$\mathcal{H}: -\frac{2}{3}x_4 + \frac{1}{3}x_2 + x_3 - \frac{4}{3} = 0$$

$$\overrightarrow{\mathcal{H}}: -\frac{2}{3}x_4 + \frac{1}{3}x_2 + x_3 = 0$$

Teste degli autovalori: $M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ $|M| = 2 \neq 0$

$$Q \in \mathcal{H} \Leftrightarrow \begin{vmatrix} 1 & 2 & x_4 - 3 \\ 2 & 1 & x_2 - 4 \\ 0 & 1 & x_3 - 2 \end{vmatrix} = 0 \Leftrightarrow x_3 - 2 + 0 + 2(x_4 - 3) - x_2 + 4 - 4x_3 + 8 = 0$$

$$\Leftrightarrow 2x_4 - x_2 - 3x_3 + 4 = 0$$

Determinare il piano $\mathcal{H}'$ parallelo a $\mathcal{H}$ e passante per $A(-2, 1, 5)$

$$\mathcal{H}': 2x_4 - x_2 - 3x_3 = 0$$

$$\mathcal{H}': 2x_4 - x_2 - 3x_3 + d = 0$$

$$A \in \mathcal{H}' \Rightarrow 2(-2) - 1 - 3 \cdot 5 + d = 0 \Rightarrow d = 20$$

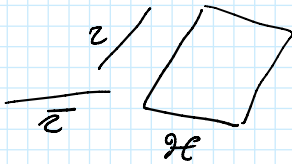
$$\mathcal{H}': 2x_4 - x_2 - 3x_3 + 20 = 0$$

Determinare una retta r parallela a $\mathcal{H}$ e passante per $B(3, 1, 0)$

$$r: \begin{cases} x_4 = 3 + t_1 \\ x_2 = 1 + 2t_1 \\ x_3 = 0 + 0 \end{cases}$$

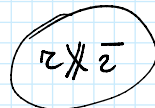
$$r: \begin{cases} t_1 = x_4 - 3 \\ x_2 = 1 + 2x_4 - 6 \\ x_3 = 0 \end{cases}$$

$$\begin{cases} x_2 - 2x_4 + 5 = 0 \\ x_3 = 0 \end{cases}$$



$$r, \bar{r} \parallel \mathcal{H} \parallel \mathcal{H}'$$

$$\bar{r}: \begin{cases} x_4 = 3 + 2t_1 \\ x_2 = 1 + 1t_1 \\ x_3 = 0 + t_1 \end{cases}$$



$(\vec{E}, \Pi)$ $\mathcal{H}, \mathcal{H}'$ sottosp. eucl. de. (o affini)

$\mathcal{H}$ e $\mathcal{H}'$ sono incidenti se $\mathcal{H} \cap \mathcal{H}' \neq \emptyset$

$\mathcal{H}$ e $\mathcal{H}'$ sono sghembi se $\mathcal{H} \cap \mathcal{H}' = \emptyset$ e $\mathcal{H} \nparallel \mathcal{H}'$.

Esercizio: $\dim E = m$ $R = (0, 1)$

Siano $P(c_1, \dots, c_m)$ e $Q(d_1, \dots, d_m)$ due punti distinti.

Determinare la retta passante per P e Q . $\overrightarrow{PQ} \neq \vec{0}$

$$P, Q \in r \quad \vec{r} \ni \overrightarrow{PQ}$$

$$\vec{r} = \mathcal{L}(\overrightarrow{PQ}(d_1 - c_1, \dots, d_m - c_m))$$

$$r: \begin{cases} x_1 = d_1 + (d_1 - c_1)t \\ x_2 = d_2 + (d_2 - c_2)t \\ \vdots \\ x_m = d_m + (d_m - c_m)t \end{cases}$$

$$\dim E = 2 \quad P(-3, 4) \quad Q(0, 1)$$

$$r: \begin{cases} x_1 = 0 + 3t \\ x_2 = 1 - 3t \end{cases}$$

$$\vec{r} = \mathcal{L}(\overrightarrow{PQ}(3, -3))$$

$$\dim E = 2 \quad P(-3, 4) \quad Q(0, 1)$$

$$r: \begin{cases} x_1 = 0 + 3t \\ x_2 = 1 - 3t \end{cases}$$

$$\vec{e} = \mathcal{L}(\overrightarrow{PQ} (3, -3))$$

$$\vec{r}: \begin{cases} t = x_1 \\ x_2 = 1 - x_1 \end{cases}$$

$$\vec{r}: x_1 + x_2 - 1 = 0$$

$$\vec{r}: \begin{cases} x_1 = 0 + t \\ x_2 = 1 - t \end{cases}$$

$$\vec{r} = \mathcal{L}(u(1, -1))$$

$$\Downarrow \\ r = \vec{r}$$

$$(\vec{E}, \mathcal{E}, \pi) \quad \dim E = 2$$

$$r: ax + by + c = 0$$

$$r': a'x + b'y + c' = 0$$

$$R = (0, 3)$$

$$\vec{r}: ax + by = 0$$

$$\vec{r}': a'x + b'y = 0$$

condizioni di incidenza
tra rette in un piano

$$r = r' \text{ opp. } r \neq r'$$

$$r \cap r': \begin{cases} ax + by + c = 0 \\ a'x + b'y + c' = 0 \end{cases}$$

$$C = \left(\begin{array}{cc|c} a & b & c \\ a' & b' & c' \end{array} \right)$$

$$1 \leq \text{rang}(A) \leq \text{rang}(C) \leq 2$$

$$\begin{matrix} 1 \\ 1 \end{matrix}$$

$$\begin{matrix} 1 \\ 2 \end{matrix}$$

$$\Rightarrow r = r'$$

$$\Rightarrow r \cap r' = \emptyset$$

$$\frac{r}{2} = \vec{r}'$$

$$\vec{r} \cap \vec{r}': \begin{cases} ax + by = 0 \\ a'x + b'y = 0 \end{cases}$$

$$2$$

$$2$$

$$\Rightarrow |r \cap r'| = 1$$

$$(\vec{E}, \mathcal{E}, \pi) \quad \dim E = 3$$

$$R = (0, 3)$$

condizioni di incidenza tra due rette

$$r: \begin{cases} ax + by + cz + d = 0 \\ a'x + b'y + c'z + d' = 0 \end{cases}$$

$$r': \begin{cases} \alpha x + \beta y + \gamma z + \delta = 0 \\ a'x + b'y + c'z + d' = 0 \end{cases}$$

$$r \cap r': \begin{cases} ax + by + \dots + d = 0 \\ a'x + \dots + d' = 0 \\ \alpha x + \dots + \delta = 0 \\ a'x + \dots + d' = 0 \end{cases}$$

$$C = \left(\begin{array}{ccc|c} a & b & c & d \\ a' & b' & c' & d' \\ \alpha & \beta & \gamma & \delta \\ a' & b' & c' & d' \end{array} \right)$$

$$\dim \vec{r} \cap \vec{r}' = \underline{0}$$

$$2 \leq \text{rang}(A) \leq \text{rang}(C) \leq 4$$

$$2$$

$$2$$

$$\Rightarrow r = r'$$

$$2$$

$$3$$

$$\Rightarrow r \cap r' = \emptyset \text{ e } \vec{r} = \vec{r}'$$

$$3$$

$$3$$

$$\Rightarrow |r \cap r'| = 1$$

$$3$$

$$4$$

$$\Rightarrow r \cap r' = \emptyset \text{ e } \vec{r} \cap \vec{r}' = \underline{0}$$

r e r' sono sghembe

Condizioni di incidenza tra una retta e un piano:

$$r: \begin{cases} ax + by + cz + d = 0 \\ a'x + b'y + c'z + d' = 0 \end{cases}$$

$$\mathcal{H}: \alpha x + \beta y + \gamma z + \delta = 0$$

$$r \cap \mathcal{H}: \begin{cases} ax + by + cz + d = 0 \\ a'x + b'y + c'z + d' = 0 \\ \alpha x + \beta y + \gamma z + \delta = 0 \end{cases}$$

$$C = \left(\begin{array}{ccc|c} a & b & c & d \\ a' & b' & c' & d' \\ \alpha & \beta & \gamma & \delta \end{array} \right)$$

$$z \cap \mathcal{H}: \begin{cases} ax + by + cz + d = 0 \\ a'x + b'y + c'z + d' = 0 \\ \alpha x + \beta y + \gamma z + \delta = 0 \end{cases}$$

$$C = \begin{pmatrix} a & b & c & d \\ a' & b' & c' & d' \\ \alpha & \beta & \gamma & \delta \end{pmatrix}$$

"A"

$$2 \leq \text{rang}(A) \leq \text{rang}(C) \leq 3$$

$$\begin{matrix} 2 & 2 \\ 2 & 3 \end{matrix}$$

$$\Rightarrow z \subseteq \mathcal{H}$$

$$\Rightarrow \vec{e} \cap \vec{\mathcal{H}} = \vec{z} \text{ e } z \cap \mathcal{H} = \emptyset$$

quindi: $z \parallel \mathcal{H}$ e $z \cap \mathcal{H} = \emptyset$

$$\vec{z} \cap \vec{\mathcal{H}}: \begin{cases} ax + by + cz = 0 \\ a'x + b'y + c'z = 0 \\ \alpha x + \beta y + \gamma z = 0 \end{cases}$$

$$\begin{matrix} 3 & 3 \end{matrix}$$

$$\Rightarrow |z \cap \mathcal{H}| = 1$$

↖ Tes. di Cramer

Condizioni di incidenza tra due piani: ESERCIZIO.

$$\dim E = 3$$

$$R = (O, B)$$

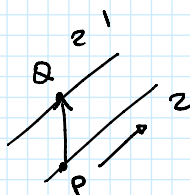
due rette parallele oppure incidenti: sono complanari; ossia esiste un piano che contiene entrambe

$$z: \begin{cases} ax + by + cz + d = 0 \\ a'x + b'y + c'z + d' = 0 \end{cases}$$

$$z': \begin{cases} \alpha x + \beta y + \gamma z + \delta = 0 \\ \alpha'x + \beta'y + \gamma'z + \delta' = 0 \end{cases}$$

$$\bullet z \parallel z'$$

$$\vec{z} = \vec{z}' = \mathcal{L}(u)$$



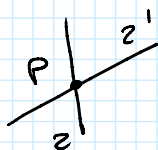
$$\mathcal{H} = (P, \mathcal{L}(u, \overrightarrow{PQ})) \supseteq (P, \mathcal{L}(u)) = z$$

$$\supseteq (Q, \mathcal{L}(u)) = z'$$

$$\bullet z \cap z' = P$$

$$\vec{z} = \mathcal{L}(u)$$

$$\vec{z}' = \mathcal{L}(v)$$



$$\mathcal{H} = (P, \mathcal{L}(u, v))$$

$$z: \begin{cases} x + y = 1 \\ y - z = 0 \end{cases}$$

$$z': \begin{cases} x + 2y = 2 \\ 3y - z = 1 \end{cases}$$

$$C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 1 & 2 & 0 & 2 \\ 0 & 3 & -1 & 1 \end{pmatrix}$$

$$|C| = \begin{vmatrix} 1 & -1 & 0 \\ 2 & 0 & 2 \\ 3 & -1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 3 & -1 & 1 \end{vmatrix} = -6 + 2 + 2 + (-1) - 1 + 3 = 2 \neq 0$$

$$z'': \begin{cases} x = 2 + t \\ y = -1 + t \\ z = -1 - t \end{cases}$$

$$\vec{z}'' = \mathcal{L}(w(1, 1, -1))$$

$$z \not\parallel z''$$

$$z \cap z'' = P(2, -1, -1)$$

$$\vec{v}: \begin{cases} x+y=0 \\ y-z=0 \end{cases} \quad S_0 = \{(-y, y, y) \mid y \in \mathbb{R}\}$$

$$\vec{v} = \mathbb{L}(\mu(-1, 1, 1))$$

$$\mathcal{H}: \begin{cases} x = 2 + t_1 - t_2 \\ y = -1 + t_1 + t_2 \\ z = -1 - t_1 + t_2 \end{cases}$$

$$\begin{vmatrix} 1 & -1 & x-2 \\ 1 & 1 & y+1 \\ -1 & 1 & z+1 \end{vmatrix} = 0.$$